

Battery Management Systems for Electric Vehicles: Architectures, Estimation Techniques, and Emerging Trends

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Abstract- Electric vehicles are increasingly utilized in the transition to sustainable transport globally. Currently, a battery management system in LIBs is predominantly needed to guarantee safe, effective, and durable performance. Nevertheless, despite significant achievements in LIB technology in terms of precise battery state estimation and effective thermal management in various conditions, it is still necessary to work towards improvement in these aspects. This paper evaluates and summarizes recent significant developments in advanced SOC and SOH estimation techniques in EV batteries. Specifically, machine learning algorithms, hybrid estimation schemes, and other types of Kalman Filters are highlighted in more detail. Also, in addition to that, there is a review in thermal management schemes that encompass integrated power-thermal systems, phase change materials, and liquid cooling systems. By carrying out an analysis, it has been proven that using an advanced technique compared to traditional ones reduces degradation and enables designers to minimize safety risks. Also, complementing intelligent control algorithms with thermal management effectively enhances EV battery lifespan and general energy efficiency. This paper has important applications and uses; it greatly assists through its investigation by presenting novel and creative ideas on electric vehicle development.

Keywords- Battery Management System, State of Charge (SOC), State of Health (SOH)

I. INTRODUCTION:

The rising need for environmentally friendly and cleaner modes of transportation is also propelling the development of Electric Vehicles. The reliability, performance, and overall safe handling of Electric Vehicles are greatly dependent upon the efficacy of a Battery Management System that monitors, protects, and manages lithium-ion batteries, as of now the most popular power storage solutions in Electric Vehicles because of their high storage density and long cycle lives. As EV adoption is gaining momentum, estimation of SOC and SOH has become critical. SOC is a measure of the remaining usable capacity, whereas SOH reflects the degradation of the battery. The initial estimation methods were based on simple electrical measurements and models, such as enhanced Coulomb-counting methodology based on relationships between SOC and OCV [1], with extensions for consumer application. However, all these approaches tend to drift and

show hysteresis, and their accuracy is reduced during dynamic driving cycles.

To overcome these challenges, model-based approaches, mainly variants of the Kalman Filter, have recently been of more pronounced interest. Papers on such studies related to EKF, UKF, and adaptive Kalman filters in the context of battery reviews deal with efficient handling of non-linear battery dynamics and noise effects associated with sensors in real-time SOC estimations effectively [2]. The latest developments on such fronts involve adaptive UKF algorithms for concurrent SOC and parameter estimations together with fractional-order Kalman filters [3][4].

At the same time, there has been development in SOH estimation through data-driven approaches, making use of voltage interval features, which provide faster and more efficient estimation [5]. Hybrid models, including co-estimation using the joint model of LSTM and AUKF based on deep learning and adaptive filtering, and twin time-scale neural models capable of providing joint estimation of SOC, SOH, and RUL, respectively, have been developed [6][7].

"Community work in other areas of computations, specifically solvers for symmetric systems and machine learning methods for SVM problems, form the basis for improvements in diagnosis functions." where the very significant position of intelligent control systems is held within EV infrastructure development [10][11], that within the comparisons of the different types of batteries, confirmation of reliability has been affirmed with lithium-ion version [12].

Contemporary research work is also focused on reinforcement learning for optimal charging tasks [13] and intricate energy management solutions, where SOC/SOH prediction solutions are highly relevant. The application of deep learning algorithms, like the LSTM-AUKF method, is also reflected to be more robust in the face of aging and loading patterns, and novel approaches in the design of the BMS in the context of the IoT solution improve real-time monitoring [14]. Their future applications would fall equally imperative in the context of the scenario based on the integration of EVs with micro-grids [15, 16].

The outline of this paper is described below. In Section 2, the architecture of the Battery Management System and its functional operations have been described. Section 3 discusses

the challenges associated with the EV Charging Station infrastructure in the Indian scenario. Section 4 and 5 comprise a detailed survey on the modern control techniques for EV power conversion systems and the latest battery capacity estimation techniques, respectively. Section 6 discusses the thermal management process in batteries and its significance to battery safety and performance. Section 7 describes the environmental and health effects associated with the wider use of EVs. Section 8 discusses the latest advancements in BMS technology and the associated research trends. Section 9 describes the future advancements and research trends in EV technology, and finally, the closing remarks of the paper.

II. BATTERY MANAGEMENT SYSTEM (BMS) ARCHITECTURE AND FUNCTIONS AND ALGORITHM

A BMS performs several vital tasks, as follows: 1. Obtaining Data: It is involved in continuous monitoring of parameters like battery temperature, voltage and current. State Determination: It calculates key states like State of Charge (SOC) and State of Health (SOH) based on collected data. Safety Protection: It protects the battery from potential hazards like deep discharge, charging, and heating. Thermal Management: It ensures that the battery operates at a desirable temperature. [7] [9]

A. Battery Management System (BMS):

Battery Management System (BMS): A Battery Management System monitors the battery's voltage, current, and temperature. It prevents any adverse conditions like overcharging, deep discharge, and heating. Verifies the proper functioning of the battery. The BMS balances the cells, regulates charging and discharge events, and reports faults or any issue. In advanced systems, it employs clever techniques such as Kalman Filters to accurately output battery data.

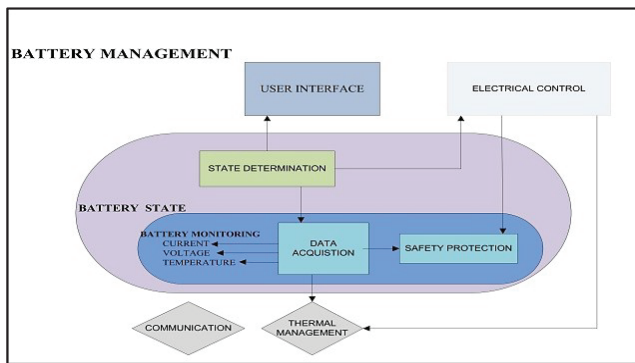


Fig. 1. Battery Management System

The quality of the BMS affects the battery's lifespan and safety. However, many systems still lack a proper State of Health (SOH) estimation. This information is crucial for understanding the battery's state and for knowing when it should be replaced [2][4].

B. SOC and SOH:

State of Charge (SOC) and State of Health (SOH) are the two most important values that are delivered by BMS [6]. SOC is the instantaneous capacity available in the battery, typically as a percentage value, with respect to the battery's maximum capacity ($\text{Max Current} \times 100\%$). Precise SOC is necessary delivering correct run-time or range estimation to the user and for governing the charge/discharge operation safely. Estimation Techniques for SOC and SOH. The estimation techniques for battery State of Charge (SOC) and

State of Health (SOH) mainly include Coulomb Counting Model- based estimation methods. The coulomb counting method- which integrates current over time, is a straightforward and rapid method. It estimates SOC by tracking the charge and discharge current over time. However, this approach has accumulated errors due to inaccuracies in current measurement and errors in initial conditions. In contrast to Model-Based Estimation methods, which are carried out through the application of the different models of the Kalman Filter algorithm, there is improved accuracy. The Kalman Filter Algorithm is utilized on many different projects and is coupled with Python programming. The application enables the prediction and estimation of the SOC value based on the dynamic movement of the battery model and sensors. The SOC and SOH estimation is highly dependent on the noise level.

The figure shows how SOC/SOH estimation accuracy relates to noise magnitude over time and illustrates how noise affects model performance.

C. BMS For SOC And SOH Algorithm:

SOH is a measure of the battery's current total charge storage capacity compared to its original, rated capacity.

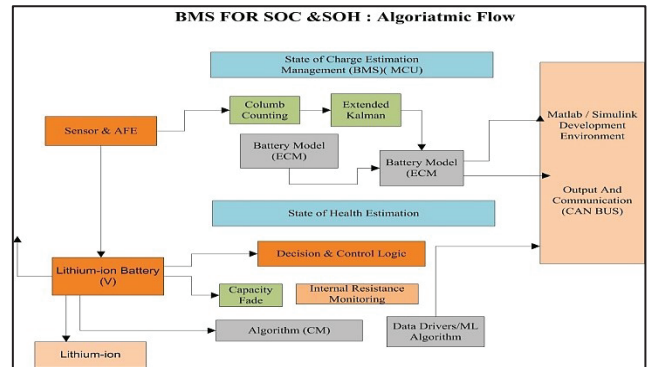


Fig. 2. BMS For SOC and SOH: Algorithm

Figure 2: Show the BMS for SOC And SOH algorithm flow of the estimation of the SOC and SOH.

III. EV CHARGING STATION CHALLENGES IN INDIA:

In India EV charging station mainly challenges due to the encompass infrastructure, technology, finance, and consumer behavior, presenting a complex issue for both government and private operators.

A. Infrastructure and Installation Hurdles:

High costs and logistical challenges hamper charging station installation. High Upfront Costs: The initial investment is huge as it includes expensive equipment, especially DC fast chargers, land acquisition, and grid upgrades like new transformers, which are often required to bear the high electrical load. Land Acquisition: Acquiring suitable locations is a major challenge. It is hard to find space in densely populated urban areas, and more so at a great cost. This is further exacerbated by bureaucratic delays and hurdles in getting "right of way" permissions. Grid Instability: Most areas of the Indian grid cannot withstand high, sudden jumps in power demand when many such fast chargers operate simultaneously. These lead to grid overloads and need expensive upstream infrastructure upgrades. Poor Location Planning: Many charging stations were installed during the early stages without proper demand analysis. They ended up

in low EV traffic areas, leading to underutilization and making the stations financially unviable.

B. Technology and Standardization Issues:

The absence of standardization in technology causes confusion and compatibility issues. Different Charging Standards: Currently, India has different charging standards for the Connector of the Charger (e.g., CCS2, CHAdeMO, and the old Bharat DC-001). A lot of electric cars will be unable to be charged by the existing station due to the standard, hence the need for different and costly connectors at one station. Obsolete Chargers: Most of the existing electric charging stations, especially the ones installed under the old government initiative, are of the old Bharat-001 standard. Most new electric models are incompatible with these, hence making the installation obsolete and redundant. Dependence on Imports: Key components for chargers, such as Battery Management Systems, popularly known as BMS, and sophisticated connectors, are usually imported. This has contributed to higher production costs and a fragile supply chain.

C. Low Utilization Rates:

The low penetration of electric cars, particularly four Wheelers, results in most charging points being underutilized. This is considered the single largest drawback for such charging points to be profitable. Slow Return on Investment (ROI): Because of primary investment costs and limited revenue that accrue due to underutilization, usually anything between 3-5 years and even longer. High Operational Costs: Apart from the investment, additional costs for electricity, software, maintenance, and even employees for running the station [10][13].

D. Awareness, Range Anxiety, and User Experience Gaps:

User Awareness Despite growth, the awareness of station location and how to optimally use them remains low in the public. Which increases with unreliable or sparse station density and with inconsistent operational status. Complicated User Experience Multiple apps, payment silos, and lack of real-time functional information contribute to a poor user experience as well as destroy the confidence that is needed for broad adoption. [3][7]

E. BMS-Related Integration Challenges:

Smart Grid and BMS Connectivity: Increasingly, advanced charging stations are now equipped with energy management systems that demand smooth interfaces with the vehicle BMS for optimum battery utilization and fast charging. Safety and Standards: A high level of monitoring standards in the charger and the BMS for prevention of thermal runaway, voltage, and current is required, and any fault involves a high level of risk.

IV. CONTROL TECHNIQUES FOR EV CONVERTERS:

The control methods employed substantially influence the performance and robustness of the converters employed in EVs, methods based on artificial intelligence.

Control methods can be classified into classical control techniques, AI control techniques. The basic control methods involve basic PI control or PID control of linear systems. Hysteresis Current Control is also a basic control method. It has a fast response. The Pulse Width Modulation (PWM) method is also a basic control method.

More advanced techniques deal with a broader level of needs. Model Predictive Control, MPC, involves the use of system models for state predictions in multivariable systems. Field Oriented Control, FOC, enhances the control of motors.

AI and optimization-based methods offer flexible solutions. Fuzzy Logic Control works for systems without precise models. Neural Networks can learn system dynamics from data. Reinforcement Learning allows controllers to discover optimal policies through trial and error for self-optimizing controls.

V. STATE OF CHARGE (SOC) ESTIMATION:

The most important factor in a Battery Management System (BMS) is the State of Charge (SOC). This parameter indicates how much energy is left in the battery. Data collection and preparation involve low battery data, process of cleaning, machine learning and algorithm data base [8]. Getting the SOC right is essential for the safe and efficient operation of electric vehicles. It helps to avoid deep discharge and overcharging, which can shorten battery life and harm performance. Several methods exist for estimating SOC, and each has its own advantages and disadvantages. Generally, these methods fall into four categories. Traditional SOC estimation methods include the ampere-hour integration method [9]. Open circuit voltage method [10]. State of Charge (SOC) estimation methods can be broadly categorized into three major approaches: Model-Based & AI Integration, Model-Based Methods, and Hybrid Measurement and Bookkeeping Methods.

A. Model-Based & AI Integration:

This is an approach which uses both mathematical models and algorithms from AI for more accuracy and adaptability while estimating SOC. The primary methods belonging to this category are:

1) Intelligent Fuzzy-Kalman Hybrid Estimation: This approach combines fuzzy logic systems and Kalman filtering to handle model uncertainties and nonlinearities within the batteries' behaviour with increased robustness and estimation precision.

2) Reinforcement-Based Optimization on Neural Networks: This technique will apply reinforcement learning algorithms to optimize the parameters of neural networks and use those to adaptively learn from system feedback to produce better predictions of SOC.

B. Model Based Method:

The Model-Based methodology adopts electrochemical or equivalent circuit models to estimate the SOC, using measurable parameter such as current, voltage and temperature.

1) EMF Method (Electromotive Force Method): The technique estimates SOC by correlating OCV with the internal electromotive force of the battery, which involves rest periods to get accurate readings.

C. Hybrid Measurement and Bookkeeping Methods:

Hybrid methods combine data-driven, model-based, and measurement-based techniques in order to achieve real-time, high-accuracy SOC estimation for dynamic operating conditions.

- 1) Data-Driven/EMF-AI Fusion: It integrates AI algorithms with EMF-based modelling to improve the accuracy of prediction by huge datasets and adapting learning.
- 2) Deep Learning from Sensor Data: Employs deep neural networks to analyse sensor data and extract complex nonlinear relationships between input variables and SOC.
- 3) Real-Time Multi-Fusion: It fuses multiple estimation sources, like model-based outputs and real-time sensor data, to enhance reliability and performance during operation.

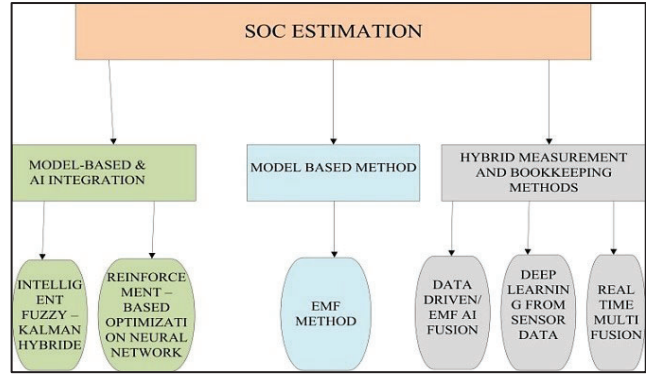


Fig. 3. SOC Estimation

Figure:3 shows the classification of SOC estimation based on model based-AI Integration having Fuzzy Kalman and reinforcement optimization on neural network, Model Based Method of EMF method and hybrid measurement which have the categorize data driven method, deep learning and real time multi fusion.

Estimating the State of Charge (SOC) accurately is crucial for battery management systems. There are four main groups of methods, each with their own, calculations, benefits, and drawbacks.

VI. THERMAL MANAGEMENT OF EV BATTERIES:

Proper thermal management is essential for the functionality, safety, endurance of Lithium-ion Batteries (LIBs).

Cooling techniques are:

- Air cooling: This type of cooling technique is quite simple and inexpensive, but it has poor heat dissipation capability
- Liquid cooling: It has better cooling control and is suitable for powerful systems.
- Refrigerant cooling and phase change materials (PCMs): These are more complex techniques that offer effective thermal stability.[11]
- Keeping the battery temperature within safe depend significantly on motor selection.

VII. IMPACT OF EV ON HEALTH AND THE ENVIRONMENT.

Though EVs lower operational emissions, the overall environmental impact depends on:

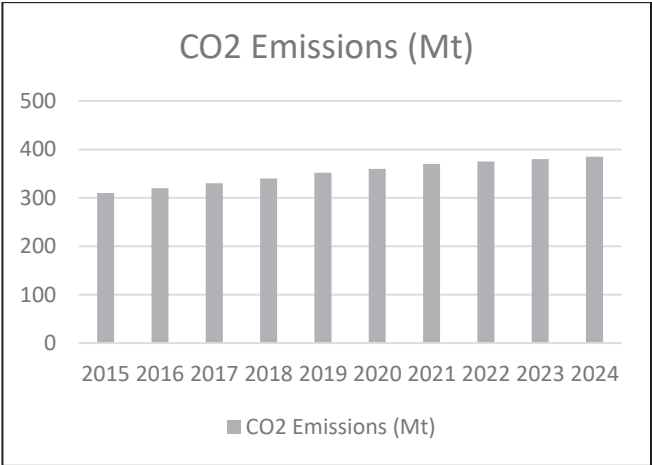
- Raw material extraction for batteries, which presents ecological and social challenges.
- Battery recycling and disposal to reduce contamination and recover materials.

- The energy source for electricity generation; renewables improve benefits, while fossil fuel-based grids limit gains.
- Following are The Environmental and Health Impact Of EV.

A. Carbon Emission:

The protection of the battery is deeply rooted in the appropriate control of the BMS, which, in turn, relies on advanced SOC and SOH estimation algorithms such as the Kalman Filter. In-depth insight and an accurate output provided by these advanced algorithms prevent overcharge, over-discharge, overheating, deep cycle, and uneven cell balancing conditions that may take place. These issues could lead to capacity degradation, inefficiency, lithium plating, or even thermal runaway, which is a concern for safety.[7][12]

The BMS can prevent these harmful effects by keeping the battery within operating limits-usually within a range of 10 to 40°C-and thus extend its life and improve reliability. Optimization is an activity that can also be counted in the arena of accomplishing the maximization of the total energy flow and the lifetime of the battery because of the extended lifetime of the environment due to zero tail pipe emissions and a low carbon footprint in the manufacturing of the battery.



Graph:1 Carbon Emission

Graph:1 reveals that the rise in carbon emissions has affected our environment, which has become highly harmful and dangerous for our society. From the data available from 2015 to 2024, the rise in carbon emissions has been substantial, but Advanced Battery Management Systems would help reduce carbon emissions.[11]

Net Zero Scenario: more than 380 million EVs, over 60% market share. This requires the quickest transition to achieve full decarbonization, with massive investment in batteries, charging, and the grid.

VIII. ELECTRIC VEHICLE BMS TECHNOLOGIES AND TRENDS

Table 1: Explained the SOC, SOH, integration and future trends, categorized in current technology, advantages and limitations.

TABLE I. ELECTRIC VEHICLE BMS TECHNOLOGIES AND TRENDS

Aspect	Current Technology	Advantages	Limitations	Advancements / Outlook	Examples
SOC	Coulomb counting, voltage and model-based methods	Simple, reliable	Accuracy drops with aging and temperature	AI/ML-based SOC estimation, adaptive algorithms	Tesla, BYD, BMW, Tata Motors
SOH	Cycle count, resistance, capacity tracking	Easy to monitor health trend	Inaccurate under dynamic use	Predictive SOH using AI, digital twin models	Tesla, Mercedes-Benz, Mahindra, Tata Motor
Integration	Microcontroller-based with basic filters	Real-time data protection	Limited processing power	AI chips, edge computing, OTA updates	Hyundai, GM, Honda
Future Trend	Hybrid AI-physics models	Better accuracy, predictive	High data and computing needs	Digital twin BMS, smart balancing, cloud AI	Tesla, Hyundai, Tata Motors

A. Challenges of EV in India:

- High purchase costs from expensive batteries.
- Limited public charging infrastructure.
- Slow regulatory approvals and inconsistent policies make rollout efforts difficult.
- The dependence on the battery minerals such as lithium, cobalt, and nickel through the supply chain causes various price and ethical concerns.
- Battery durability, environment, and recyclability have been raised as considerations.
- The initial investment for EVs increases their cost by 20-30 percent compared with traditional vehicles.
- reliability of power supplies in the region can pose challenges to the functionality of charging stations.

These challenges need coordinated solutions, such as improved infrastructure, stable policies, local manufacturing, and greater consumer awareness, accelerate EV growth in India and worldwide.[5][13].

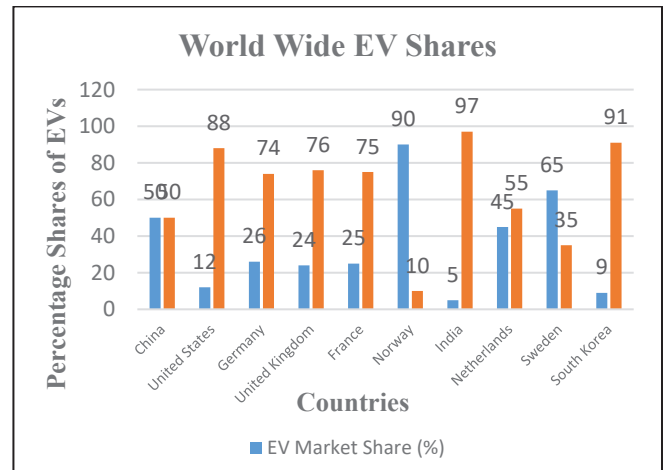
IX. FUTURE SCOPE OF EV VEHICLES:

Conservation Scenario: About 245–300 million EVs worldwide, reach 50–55% market share; growth is steady yet slower amid moderate policy and infrastructure development.

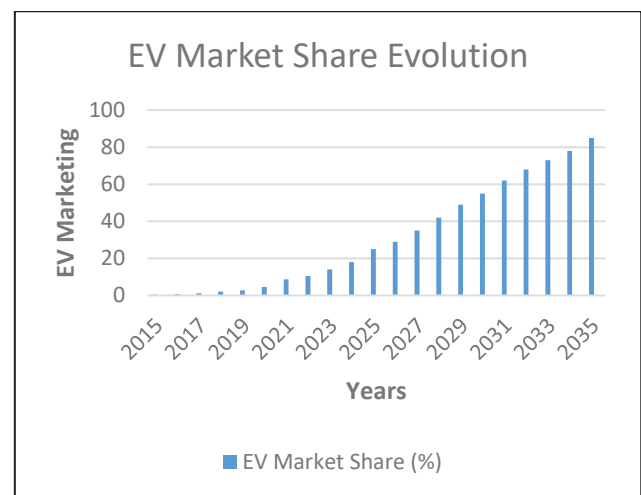
The Graph:2 represents worldwide vehicle data, where Electric Vehicles (EVs) are shown in sky blue color and traditional Internal Combustion Engine (ICE) vehicles are shown in red color.

Graph:3 presents the global trends of EV adoption. It shows how the world is inching its way to the APS, under which about 300–380 million EVs will be on the road to attain a market share of 55–60%, partly based on much firmer policy support and also probably due to rapidly declining battery costs [5, 7]. At this level of penetration, the need for highly accurate BMS becomes critical, especially for proper SOC

and SOH estimation aimed at safety, long ranges, and effective use of the battery.



Graph:2 Worldwide Shares



Graph: 3 EV Market Worldwide Exaptation

For comparison, the Net Zero Scenario forecasts more than 380 million EVs and over 60% market share, representing the most rapid route to full decarbonization. This will involve an unprecedented amount of investment worldwide for manufacturing batteries, installing charging networks, and reinforcing the grid. Such a speed further enhances the importance of using SOC/SOH estimation algorithms capable of sustaining battery reliability and lifecycle performance for large-scale energy management of fleets of EVs in the future.

X. CONCLUSION:

The study above gives insight into the important role of battery management systems in EV development. To achieve optimal performance and longevity in batteries, accurate SOC and SOH computation, efficient thermal management systems, and reliable safety systems are essential. Artificial intelligence and machine learning have created new paradigms to enhance current state estimation calculation techniques.

Systems thinking methodology integrates various systems such as electrical, thermal, and mechanics to satisfy EV demands in the future. It will be crucial in harnessing the full potential of EV in serving as a base for sustainable mobility in addressing cost and development constraints in using EV.

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